

COMS20006

SOFTWARE ENGINEERING PROJECT

AWS Architecture and Accounts

Jordan Taylor
Computer Science Teaching Technologist
jordan.taylor@bristol.ac.uk

bristol.ac.uk

AWS Architecture

Structure:

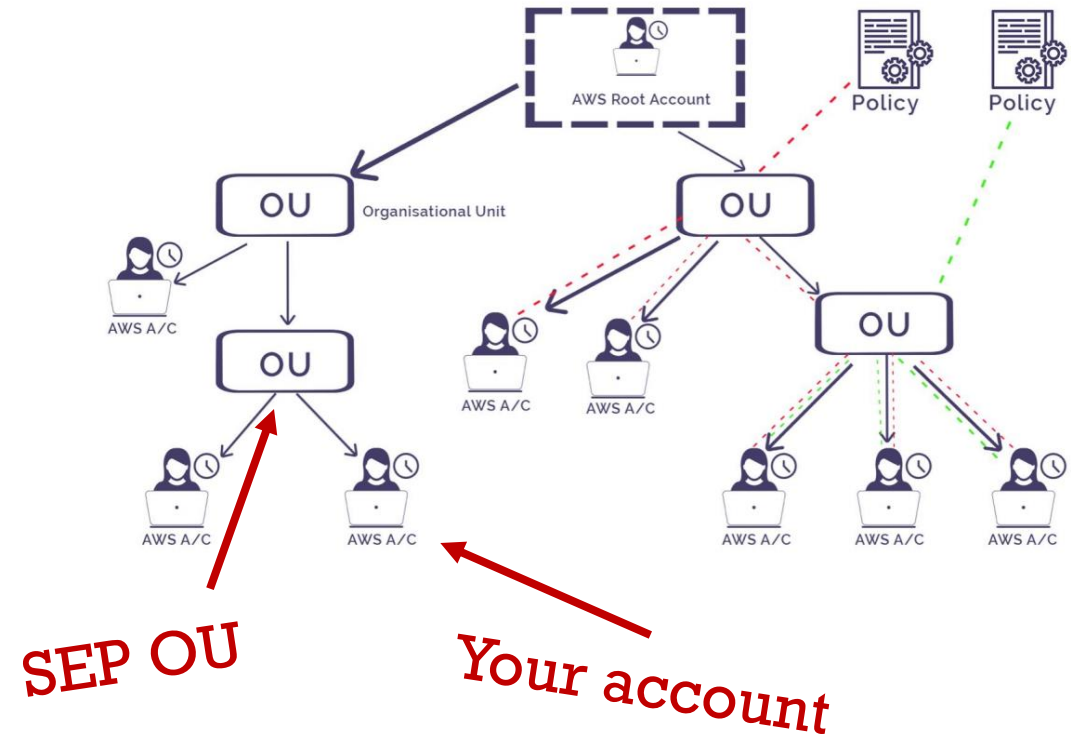
The AWS Environment is managed through **AWS Organisations** and **Control Tower**.

It is a hierarchical structure, with **Service Control Policies** governing the ability of student **Accounts** to access services and resources.

All **Accounts** reside in **Organisational Units (OUs)**.

This is what **we** see.

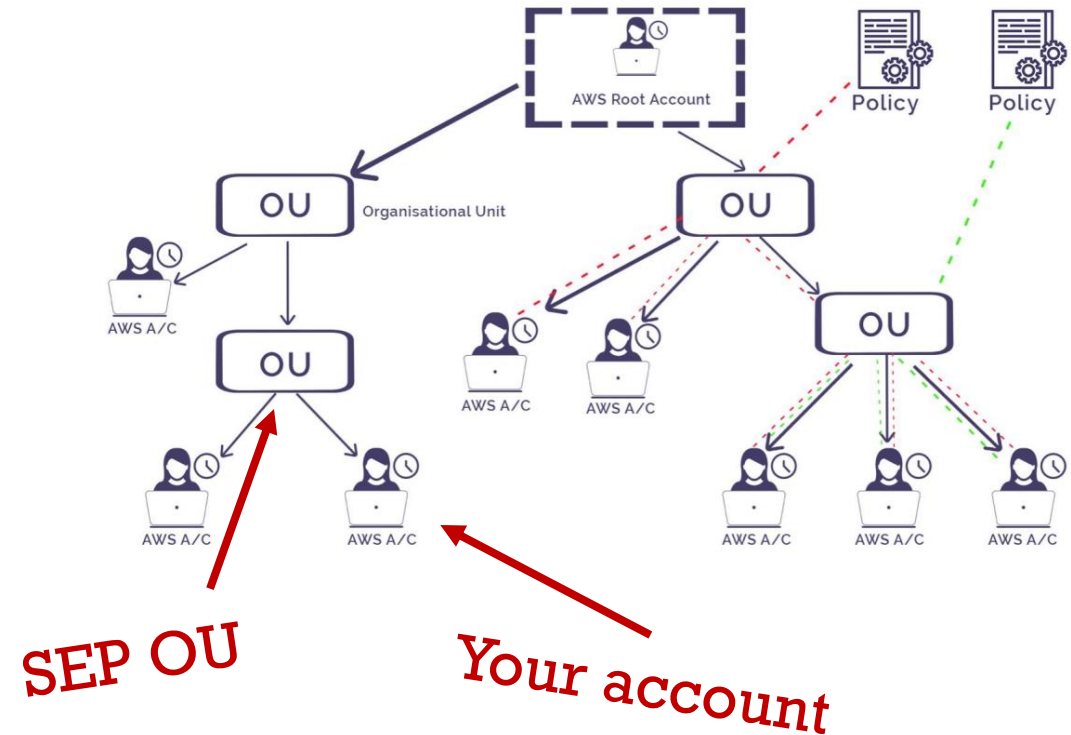
Your **Account** is a leaf in the tree.



AWS Architecture

Structure:

You will have almost complete freedom *within* your **Account**, governed by administrator-set **Service Control Policies**.



AWS Architecture

Budget:

You will have a **budget** set to your **Account**.

This **budget** is monitored on our end.

Note that exceeding the budget will result in restrictions placed on your **Account**, but will *not* immediately delete your work.

You *must* contact an administrator **and** Unit Director if you exceed your budget.



AWS Architecture

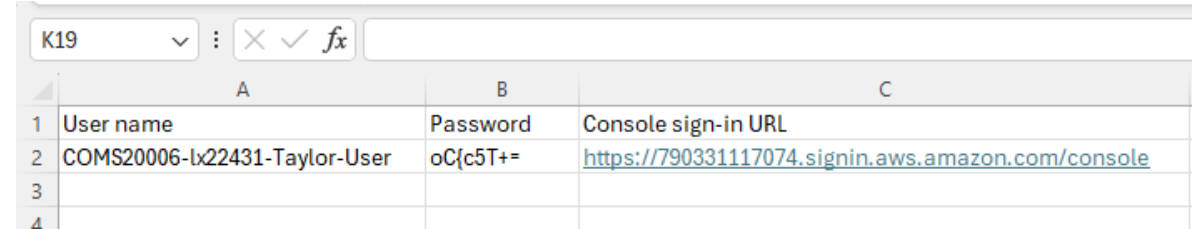
User Credentials:

When your account is provisioned, a nominated member of your team will be provided with a credentials file.

This file is for **Console** access – not for programmatic access via SDK.

The credentials file contains a **sign-in URL**, **username**, and **password**.

These are **User** credentials. Your **Account** may have multiple **Users**.



A screenshot of the AWS IAM console showing a table of user credentials. The table has three columns: 'User name', 'Password', and 'Console sign-in URL'. The first row shows a user named 'COMS20006-lx22431-Taylor-User' with a password 'oC{c5T+=' and a console sign-in URL 'https://790331117074.signin.aws.amazon.com/console'. The table is displayed in a web browser window with a search bar at the top.

	A	B	C
1	User name	Password	Console sign-in URL
2	COMS20006-lx22431-Taylor-User	oC{c5T+='	https://790331117074.signin.aws.amazon.com/console
3			
4			

*These are not genuine
credentials!*

AWS Architecture

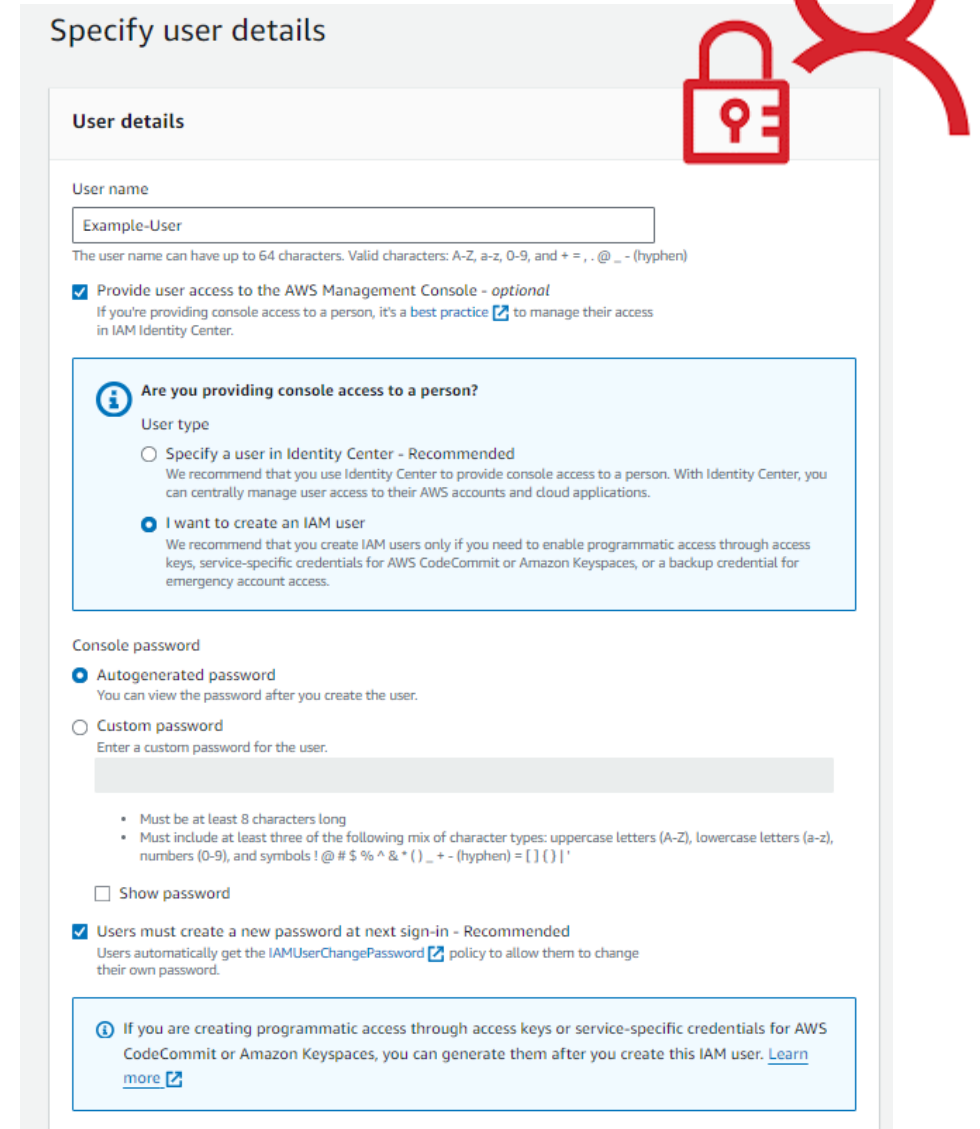
User Credentials:

It is the nominated **User's** responsibility to set up **User** credentials for the rest of the team.

This is done through the **IAM** service.

Ensure that you provision your teammates with admin policies.

Although we have **Root** access to your **Account**, you must provide the Unit Director and your Mentor with credentials.



The image shows a screenshot of the AWS IAM console 'Specify user details' page. A red padlock icon is placed over the 'User details' section header, and a red magnifying glass icon is positioned over the 'User name' input field. The page content includes:

- User details**
 - User name**: A text input field containing 'Example-User'. Below it, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)'.
 - ☒ **Provide user access to the AWS Management Console - optional**
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.
- Are you providing console access to a person?** (Information icon)
 - User type**
 - ☐ **Specify a user in Identity Center - Recommended**
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.
 - ☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.
- Console password**
 - ☒ **Autogenerated password**
You can view the password after you create the user.
 - ☐ **Custom password**
Enter a custom password for the user.
 - Must be at least 8 characters long
 - Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' "
 - ☐ **Show password**
 - ☒ **Users must create a new password at next sign-in - Recommended**
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.
- Information box**: If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Notes and Procedures

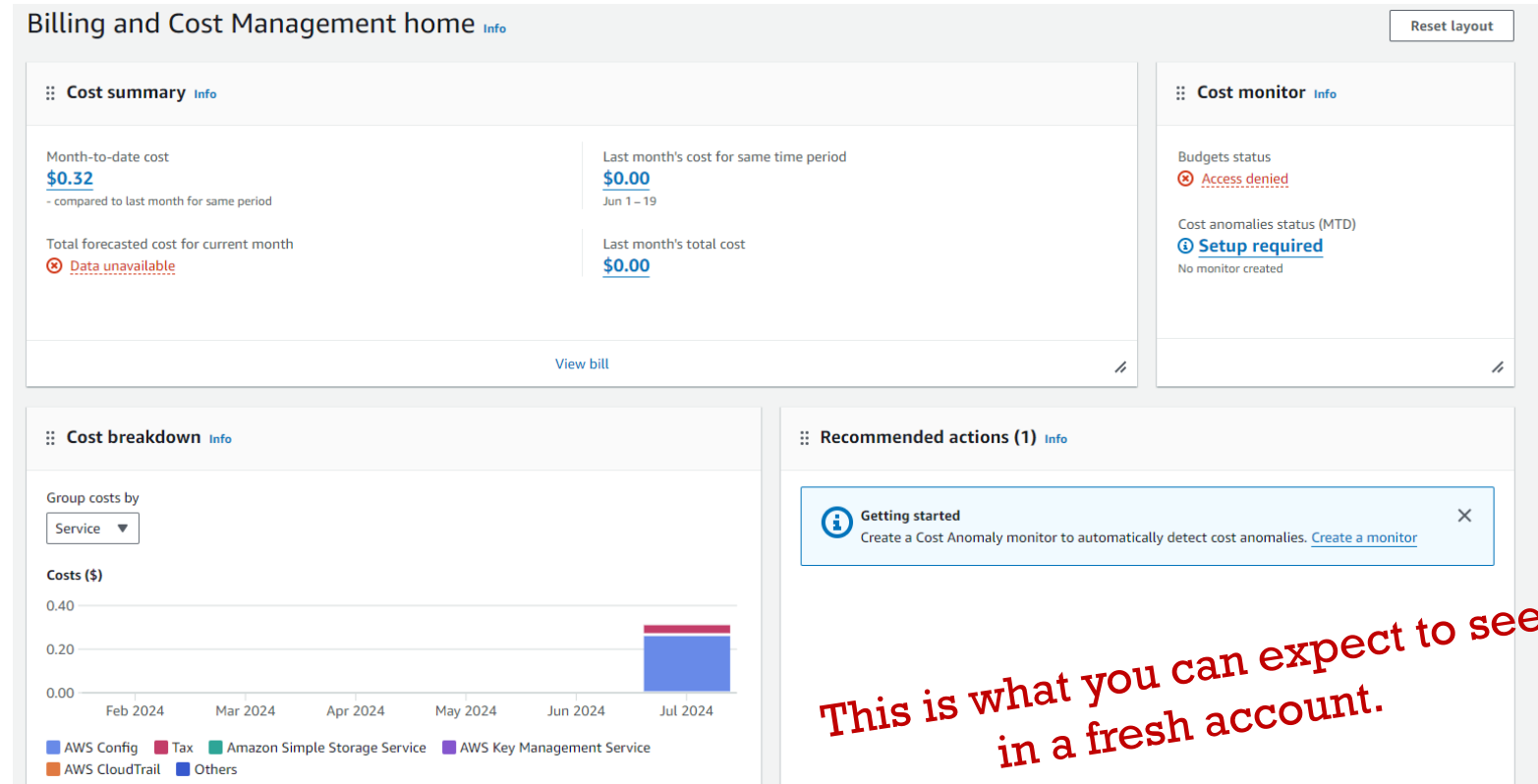
Billing

Billing and Cost Management is available through the **AWS Console**.

Almost all activities cost money.

Remember! This is **real** money you are playing with – not *AmazonFunBucks®*.

Do not, at any point, enter your own credit/debit card information.



This is what you can expect to see in a fresh account.

Note the \$0.32 used for AWS config. Don't worry about that!

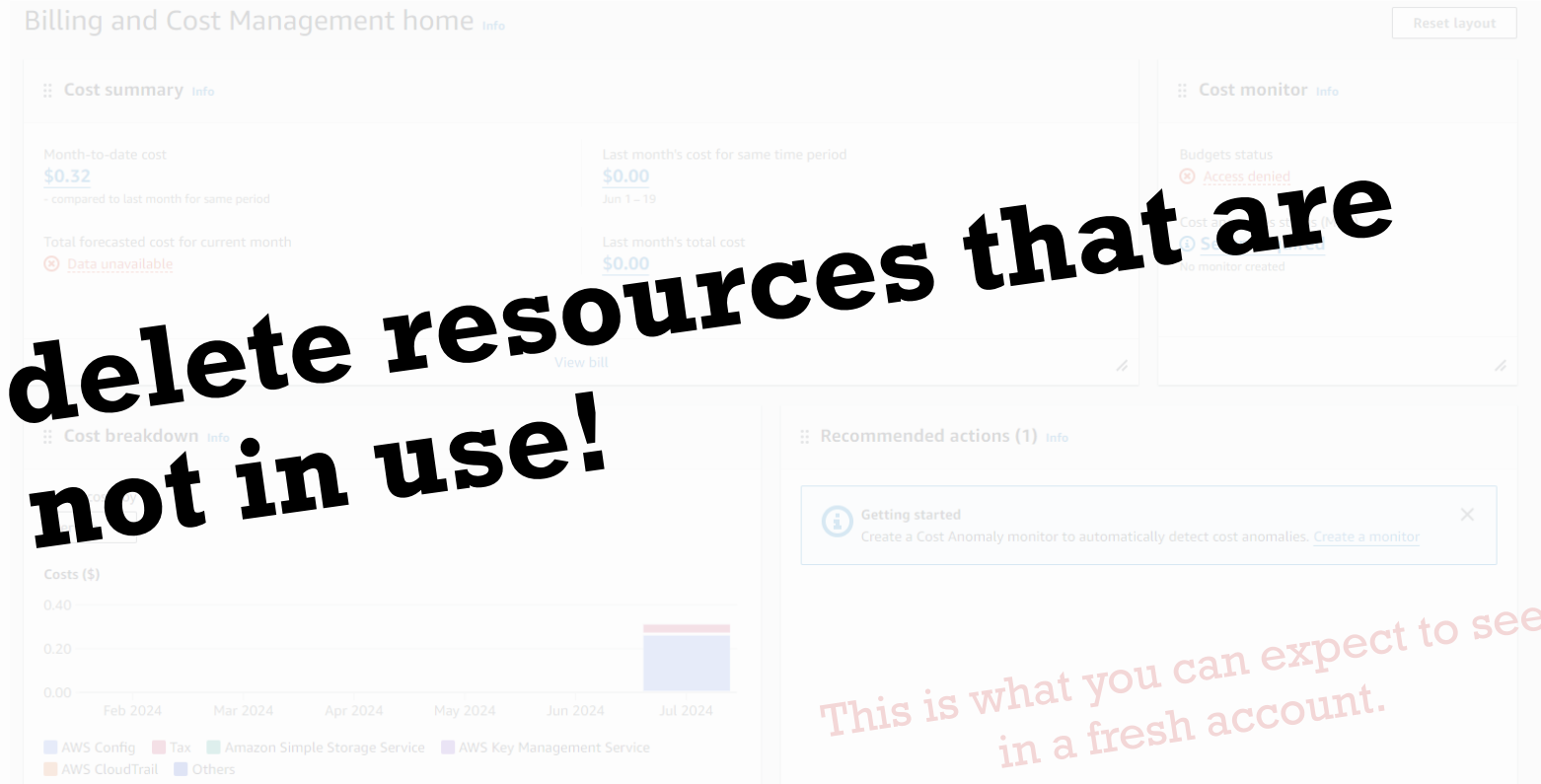
Billing

Billing and Cost Management is available through the **AWS Console**.

Almost all activities cost money.

Remember to delete resources that are not in use!

Do not, at any point, enter your own credit/debit card information.



This is what you can expect to see in a fresh account.

Note the \$0.32 used for AWS config. Don't worry about that!

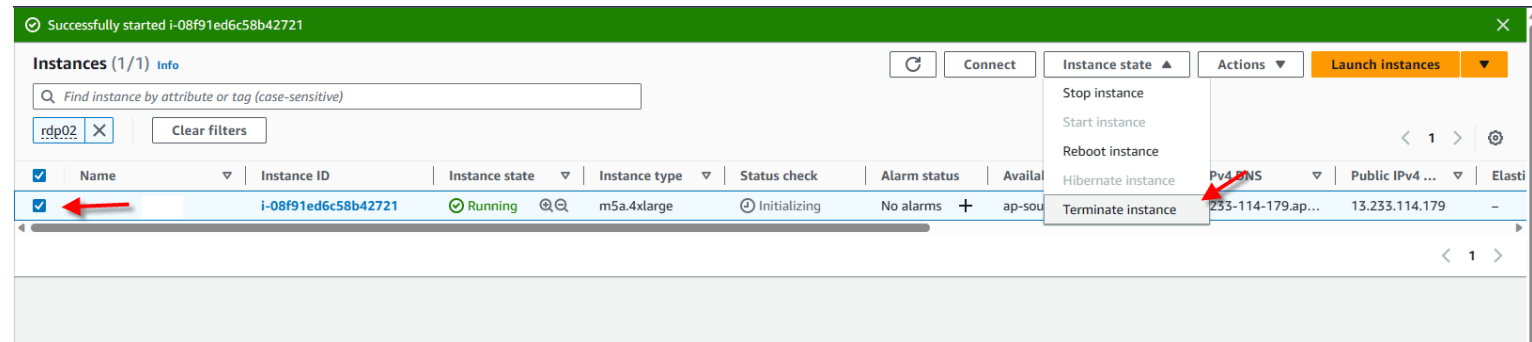
Deleting Resources

Each resource has a different method of being deleted.

One of the most common mistakes leading to accidental spend is leaving **EC2 Instances** running.

It is possible to **stop** or **terminate** an **EC2**.

Stopping an **EC2** effectively shuts it down – but you are still billed for persistent storage!



Deleting Resources

If you want to make sure that all billable resources are removed from your **Account**, I have created a **Cloudshell** tool, derived from the open-source *aws-nuke*:

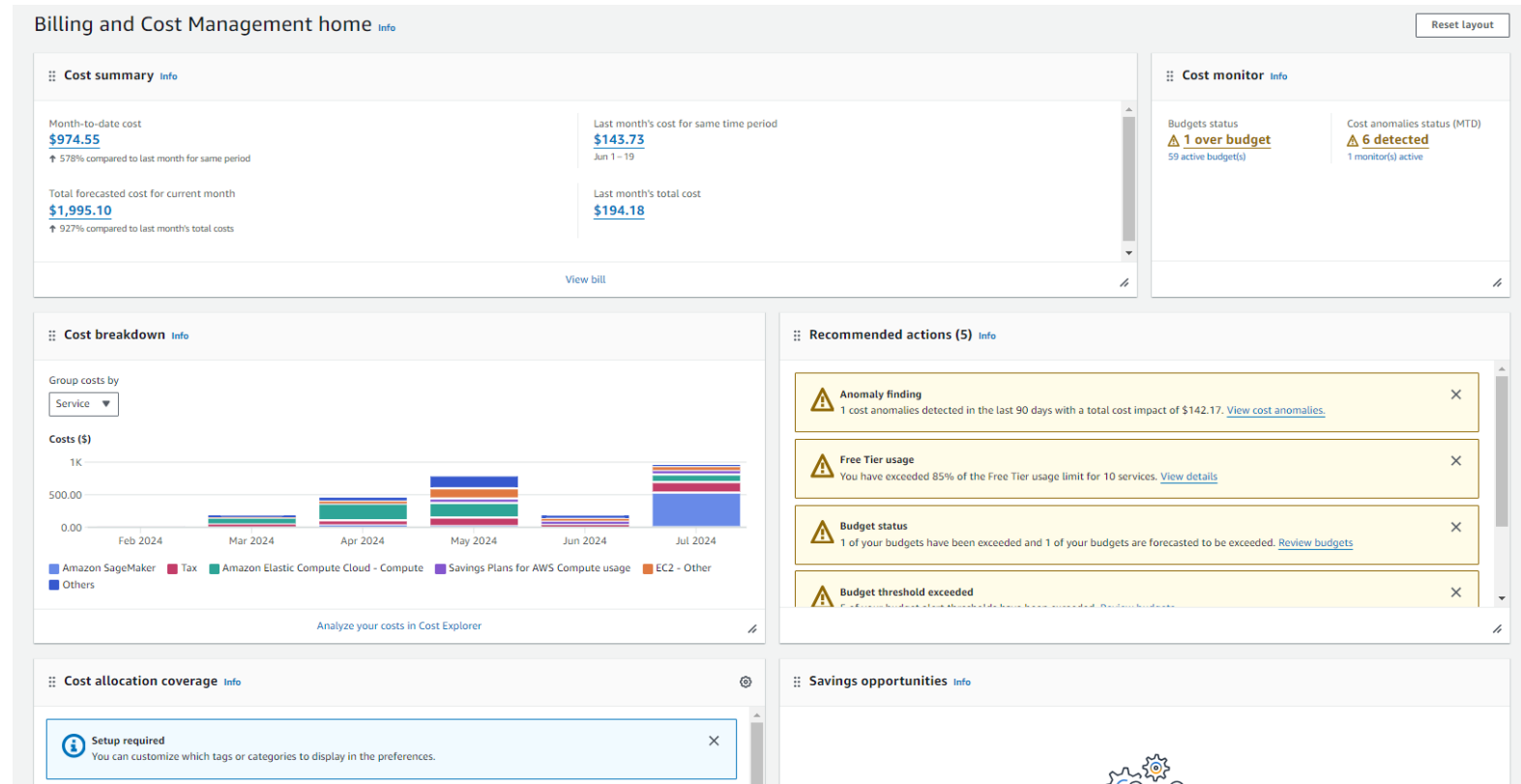
AWS BLAST

```
bash <(curl -s https://raw.githubusercontent.com/cynical-cheddar/UsefulScripts/main/blast.sh)
```

Simply paste in the above line to **Cloudshell** and it will attempt to remove **ALL** billable resources from your **Account**.

Billing

This is the admin billing panel

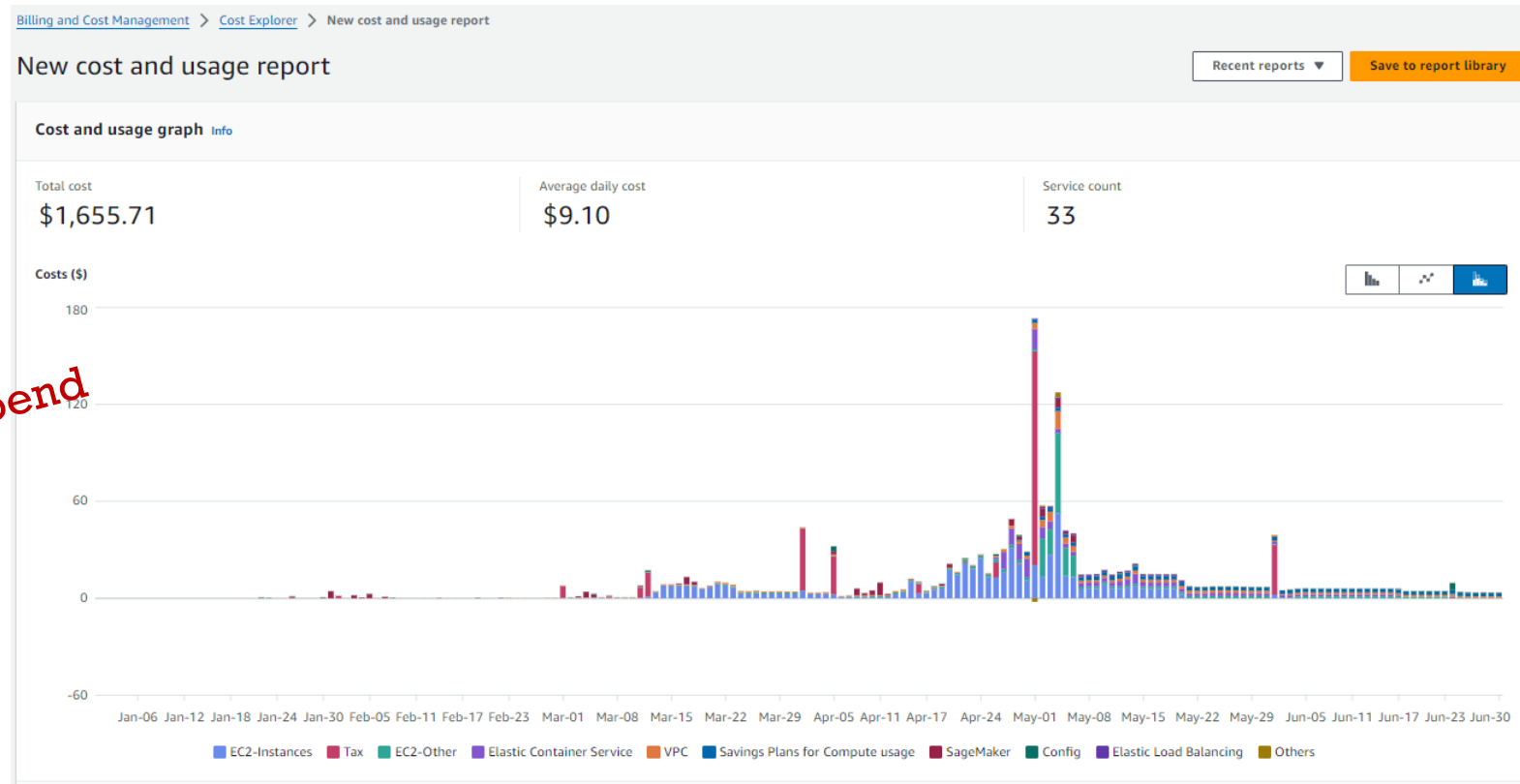


We can see everything

Billing

This is the admin billing panel

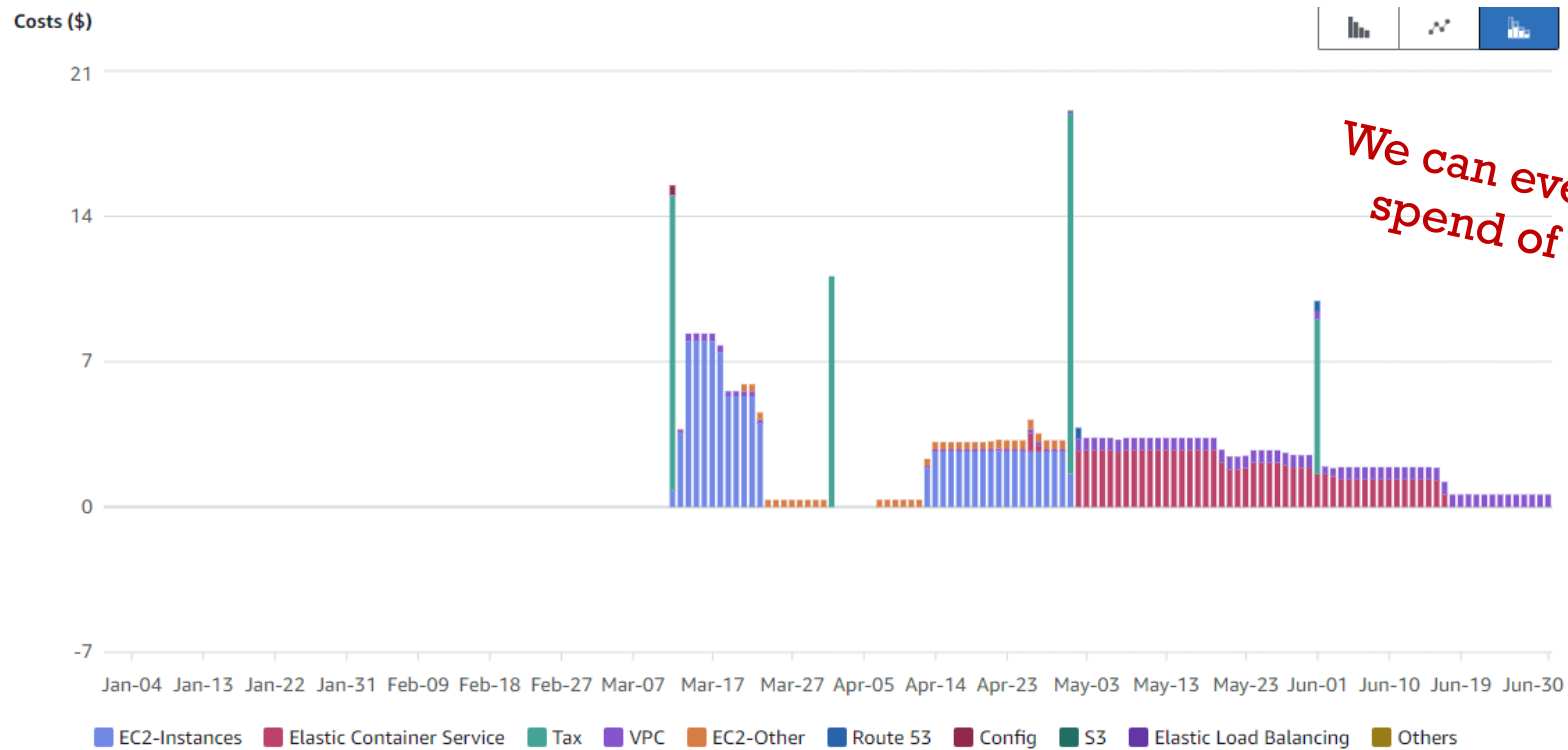
Overall Spend



We can see everything

Billing

This is the admin billing panel



We can see everything

Billing

This is the admin billing panel

So be RESPONSIBLE with your resources!



We can see everything